Fiscal Year:	FY 2015	**Task Last Updated:**	FY 08/06/2014
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2012	**End Date:**	02/28/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2017 per NSSC information (original end date was 9/30/2016)--Ed., 4/23/13		
Key Personnel Changes/Previous PI:	none.		
COI Name (Institution):	Didion, Jeffrey (NASA Goddard Space Flight Center)		
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	Electrically driven flows can provide significant enhancements to current and future technologies that can benefit from increased heat transfer and mass transport capacities in macro and micro scales. In particular, electrically generated flows (with and without phase change) have great potential for terrestrial and micro-gravity applications. Electrohydrodynamic (EHD) ion-drag pumping and induction pumping have been extensively studied in the past several decades with no significant, successful applications. On the contrary, EHD pumping based on conduction phenomenon has reached a stage where various applications are expected in the near future, since pressure heads on the order of kilo Pascal per electrode pair are easily obtainable. EHD conduction pumps are simple in design, lightweight, non-mechanical, and free of vibration and noise, and they allow for active control of heat transfer and mass transport. EHD conduction pumping requires minimal electric power to operate (at least three orders of magnitude less power		

Task Description:	compared to the resultant heat transport capacity). This proposed research will fundamentally investigate the electrically-driven dielectric liquid film in the presence of phase change in macro- and micro-scales based on conduction phenomenon through a combined experimental and computational investigation. Furthermore, the coupling of electric double layers and heterocharge layers in micro-channels will be explored, and the non-dimensional threshold at which the double layer starts to affect the corresponding flow field will be determined. Note that in micro-scales, the Debye length within dielectric fluids becomes comparable to the conduit dimensions. Additionally, unique two-phase heat transport devices will be developed that will operate based on the conduction phenomenon, with or without capillary effects, in macro- and micro-scales. Heat flux levels with these active micro-scale heat transport devices are expected to significantly exceed the corresponding working fluid critical heat flux level while operating stably.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are a number of potential terrestrial and space applications for this EHD driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure etc.). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices. For further technical and general information, please contact Professor Jamal Yagoobi at jyagoobi@wpi.edu . This device currently is not commercially available.
	The following tasks have been accomplished during the period of September 2013 to August 2014. A- Electrically Driven Liquid Film Boiling Heat Transfer in the Absence of Gravity – Variable Gravity Flight: In September 2013, the PI (Yagoobi) and Co-PI (Didion) along with two other colleagues (PhD student Viral Patel of WPI and Franklin Robinson of NASA Goddard) conducted the electrically driven liquid film boiling heat transfer experiment on board a variable gravity flight. This experiment was to be used as a pre-ISS study in preparation for future orbital lab testing. The purpose was to study the effect of gravitational acceleration on the heat and mass transfer performance of an EHD conduction pumped device. The primary objective of the variable gravity parabolic flight testing of the EHD pumped two-phase heat transfer experiment was to develop fundamental understanding of electrically driven dielectric liquid film flow in micro-scales in the absence of gravity with and without phase change (liquid/vapor). This objective was achieved through the following tasks: 1.) Demonstration of single- and two-phase operation of an EHD conduction pumped device in a low gravity environment. 2.) Study of the effects of gravity and applied EHD voltage on the boiling heat transfer coefficients achieved on a chip-scale heated surface. The radial heat transport device was a circular 1 mm thick ceramic (96% alumina substrate) disc with a high heat flux source at the center. PdAg electrodes were lithographically fabricated on the surface of the disk, forming a number of concentric circles. These electrodes used EHD conduction to pump a thin (~1 mm) liquid film over the surface of the disc from the outer periphery toward the center where the heater was installed. Bus lines for electrical connections to the electrodes were also printed on the disc. The heater at the center was formed by a 1 cm × 1 cm, 100 O resistor printed onto the bottom surface of the substrate. A 1.3 mm × 1.3 mm thermistor was printed over the heating resistor with an electrical insulation layer in between for heater surface temperature measurement. The electrodes, heater, thermistor, and associated bus lines were all printed using thick film fabrication methodology. These parts were manufactured by Thick Film Solutions. The experiment housing was a circular aluminum tank which had a glass lid and two chambers. The top chamber, in which the heat transport disc was placed, contained the working fluid, refrigerant HCFC-123. The working fluid was evaporated, pumped, and condensed in the top chamber, with a constant saturation pressure. The bottom chamber contained cooling water from a recirculating chiller and was completely isolated from the top chamber by an interface plate sealed with several o-rings. Heat was dissipated through condensation at the periphery of the disk to the chilled water. Therefore the heat exchange fluids never mixed as both chambers were perfectly sealed. The high voltage and ground feedthroughs were used to connect the electrodes on the disc surface to the high voltage power supply (discussed in next section). The high voltage wire was connected to the solder pad and common bus line of the wide electrodes, while the ground wire was connected to the common bus line of the narrow electrodes. High current-carrying wires were used for the heater power connections and this wire pair also had a separate feedthrough. Smaller low-voltage wires were used for the thermistor connections on the heater surface which were then soldered to the bus lines. As mentioned above, chambers for the water flow and HCFC-123 were isolated. The water chamber had an inlet and outlet which were connected to stainless steel tubes to and from a recirculating chiller. This cooling water provided a sink for the condensation of the HCFC-123 in the top chamber. The HCFC-123 was boiled on the heater surface and the EHD electrodes were energized to pump the film of liquid from the periphery of the disc towards the heater at the center. The HCFC-123 vapor condensed on the disc surface due to the heat loss to the flowing water in the chamber underneath. The following measurements were recorded during testing: Temperature from thermistors and RTDs (±0.05°C), absolute pressure of the fluid (Druck PDCR 4361, ± 710 Pa), acceleration due to gravity (Honeywell MA312, ± 0.02 g), applied EHD pump voltage (Trek 677B, ± 2 Vdc), and pump current (Trek 677B, ± 5 μA). An Accel 250 LC cooling/heating recirculating chiller from Thermo Scientific controlled the condenser temperature of the experiment housing. A TDK Lambda GEN150-10 power supply provided power to the heater located at the center of the electrode disc. Data was acquired using the National Instruments USB Compact DAQ 9188 system. A National Instruments (NI) GPIB-USB-HS controller connected to a notebook computer running NI LabVIEW virtual instrument software allowed users to control the loop. Following the September 2013 variable gravity flight experiments, the data were analyzed. Select results are given below. The heater surface temperatures during liquid film boiling were compared in the presence and absence of gravity, in the presence and absence of applied EHD voltage. The terrestrial data (1-g) showed that EHD conduction pumping of the liquid film enhances the heat transfer and lowers the heater surface temperature. When the EHD pump is activated, cool liquid from the periphery of the heater is continuously being transported to the

Task Progress:	heater surface. The convective heat transfer during liquid film boiling increases as a result, due to higher fluid circulation. This is true for all heat fluxes considered here. However, the results also indicate that the heater surface temperature for both terrestrial tests (1-g) was higher than those for the flight tests (0-g and 1.8-g). This was expected for the 1.8-g tests because the larger gravitational body force present significantly improved the boiling process. However, the 0-g test performed without the EHD pump activated was expected to show poorest heat transfer or highest surface temperatures, since neither the gravitational body force nor electrical body force were present to facilitate liquid film boiling. This was not the case and the reason is as follows. During the parabolic maneuver, the time spent in 1.8-g before and after every period of 0-g appeared to determine the conditions of the liquid film boiling process. Visual observations also confirmed that the liquid was forced to the surface suddenly by the higher g-force and the system took a considerable amount of time to transition to 0-g before the next 1.8-g phase started again. Therefore, the 0-g tests were more similar to the 1.8-g tests and the time to reach steady state was not adequate. Although the surface temperatures were lower (for 0-g, no-EHD), we believe that in a continuous 0-g environment, tests without the EHD pump activated will certainly result in the highest temperatures for a given heat flux, compared to terrestrial results. In fact, it is important to recognize that liquid film boiling cannot be sustained in the absence of gravity (no hydrostatic pressure) unless one can generate liquid film flow via some mechanisms. This actually justifies the need for the ISS environment versus the variable gravity flight environment that is not suitable to the proposed project as previously addressed. During the flight, at low heat fluxes (below 5W/cm2) the nucleate boiling had not initiated and the heater surface temperature was actually higher with the EHD pump activated. The reason for this was the lower convective heat transfer coefficient in this region. As nucleate boiling was initiated, bubble formation at the heater surface began and dramatically improved heat transfer, thereby lowering the surface temperature. In this region the activation of the EHD pump did indeed enhance the heat transfer and this led to lower surface temperature. The effect was diminished at higher heat fluxes as nucleate boiling dominated. The following are the conclusions of the experiments aboard the reduced gravity flight: 1. Despite generally harsh conditions (large vibrations during flight, cycling of aircraft cabin pressure and temperature etc.) compared to the terrestrial laboratory environment, the Radial Two Phase EHD Conduction Driven Heat Transport Device experiment functioned reliably. 2. The results indicate that EHD conduction pumping did successfully enhance liquid-film boiling heat transfer at moderate heat fluxes in the presence and absence of gravity. 3. The unsteady nature of the parabolic flight meant that the measured parameters did not approach steady state. Therefore all reported results are transient. From terrestrial experiments in liquid-film boiling heat transfer it was found that the required time constant for achieving steady state was ~300 seconds. Even though the total reduced-gravity flight time was 2 hours, the time spent in microgravity per parabola was only ~20 seconds which was not enough for steady-state conditions. 4. The time spent in the high-g phase of the flight (the 1.8-g phase that preceded and followed every phase of microgravity) also affected the results significantly. The large body-forces present during these phases caused liquid to “wash over” the surface and immediately lower the heater surface temperature. The system did not have enough time to recover from this phase before the next microgravity condition was reached. 5. In order to make a true comparison to earth-gravity terrestrial data, several minutes of microgravity-only conditions are needed for each data point. The only possible way that all these requirements can be met is through future experiments in continuous orbit on board the ISS Research Laboratory. B- Electrohydodynamic (EHD) Project Science Concept Review (SCR): A detailed Science Review Document was prepared in June 2014 and was forwarded to the project manager, Dr. Mohammad M. Hasan, accordingly. The Science Committee Review (SCR) for the proposed International Space Station (ISS) experiment, entitled, "Electrically Driven Film Boiling in the Absence of Gravity", was held on July 10-11, 2014. The SCR is a NASA Headquarters-led review for ISS research projects that reviews the scientific merit of the proposed investigation and the significance of the science objectives. The proposed experiment focuses on an innovative solution to transport a liquid film and enhance boiling heat transfer rates in microgravity using electrohydrodynamic conduction pumping of the dielectric working fluid. Professor Cila Herman from the Johns Hopkins University chaired the science review team consisting of Professors Markus Zahn, Massachusetts Institute of Technology; Chung Chen, University of Missouri; and Dr. Tung Lam of the Aerospace Corporation. The SCR Science Panel strongly recommended that NASA support the project, in particular the continued development of the hardware and the operation of this ISS experiment.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Articles in Peer-reviewed Journals	Patel VK, Robinson F, Seyed-Yagoobi J, Didion J. "Terrestrial and micro-gravity experimental study of micro-scale heat transport device driven by electrohydrodynamic conduction pumping." IEEE Transactions on Industry Applications. 2013 Nov-Dec;49(6): 2397-401. http://dx.doi.org/10.1109/TIA.2013.2264042 , Nov-2013
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